

SDR# Local Detection Pipeline

Overview

This pipeline runs entirely on the same Windows mini PC:

- **SDR#** tunes the receiver and outputs demodulated USB audio.
- **VB-CABLE** exposes that audio as a Windows recording device.
- **meteor-station-sdrsharp-detect** reads that local audio input.
- The detector writes event logs and review artifacts to disk on the same machine.

There is no **rtl_tcp**, no network hop, and no Raspberry Pi in this workflow.

Prerequisites

- Windows on the LattePanda Alpha
- Python 3.10 or newer
- SDR#
- VB-CABLE installed and visible in Windows audio devices
- Python dependencies installed from this repo:

```
python -m pip install -e .
```

One-Time Setup

1. Configure SDR#

Use the tuned GRAVES assumptions from the original SDR# workflow:

- Mode: **USB**
- RF tuned frequency / VFO: **143.048400 MHz**
- Expected carrier: **143.050000 MHz**
- Audio offset of interest: about **1600 Hz**
- Bandwidth: **3.000 kHz**

The detector default profile **graves_sdrsharp** is set up for that audio band and trigger behavior.

2. Configure VB-CABLE

- In SDR#, route audio output to the VB-CABLE playback device.
- In Windows, confirm the matching VB-CABLE recording/input device exists.
- Keep the Windows audio path at **48000 Hz** if possible.

3. List usable input devices

Run:

```
meteor-station-sdrsharp-detect --list-audio-devices
```

Find the VB-CABLE input device. Record either:

- its numeric device index, or
- a stable substring from its name, such as **CABLE Output**

4. Configure `meteor_station.toml`

Copy the example if needed:

```
Copy-Item meteor_station.example.toml meteor_station.toml
```

Relevant sections:

```
[audio_inputs.sdrsharp_vb_cable]
device_index = 20
device_name_contains = "CABLE Output"
sample_rate = 48000
channels = 1
block_size = 4096
dtype = "float32"
queue_max_blocks = 128

[runtime.local_sdrsharp]
output_dir = "meteor_logs"
log_level = "INFO"
save_spectrogram = true
save_wav = true
save_detection_waterfall = true
waterfall_path = "live_waterfall.png"

[detector_profiles.graves_sdrsharp]
detection_min_hz = 1200
detection_max_hz = 1600
trigger_db_above_baseline = 11.0
band_rise_db_min = 3.0
peak_to_median_db_min = 6.5
max_near_peak_bins = 5
```

Notes:

- `device_index` takes priority over `device_name_contains`.
- `waterfall_path` is relative to `output_dir` unless you provide an absolute path.
- `block_size` and `sample_rate` should normally stay at 4096 and 48000.

Daily Use

Start live detection

```
meteor-station-sdrsharp-detect --config meteor_station.toml
```

Typical startup output shows:

- live vs WAV mode
- sample rate
- block size
- output directory
- detection band
- artifact flags
- selected input device

Override settings from the command line

Examples:

```
meteor-station-sdrsharp-detect --config meteor_station.toml --device-index 18
meteor-station-sdrsharp-detect --config meteor_station.toml --device-name "CABLE
Output"
meteor-station-sdrsharp-detect --config meteor_station.toml --output-dir
D:\meteor_logs
meteor-station-sdrsharp-detect --config meteor_station.toml --no-wav
```

Stop the detector

Press **Ctrl+C**.

The detector will stop the input stream and finalize any pending event state before exit.

Output Files

By default outputs go to `meteor_logs/`.

Expected files:

- `events_v3.csv`: structured event log
- `event_v3_00001.png`: saved spectrogram for a meteor candidate
- `event_v3_00001.wav`: saved review audio for a meteor candidate
- `live_waterfall.png`: rolling waterfall image
- `waterfalls/event_v3_00001.png`: candidate review waterfall snapshot

Only `meteor_candidate` events create review artifacts. Rejected events still appear in the CSV.

Offline Validation with a WAV File

You can validate the exact detector path without SDR# running:

```
meteor-station-sdrsharp-detect --config meteor_station.toml --input-wav  
C:\path\to\fixture.wav
```

Use this to:

- verify config changes against a known recording
- test artifact generation
- compare false positives and threshold changes

The WAV file must match the configured sample rate, normally **48000 Hz**.

Troubleshooting

No audio arriving

- Confirm SDR# is sending audio to VB-CABLE, not your speakers.
- Confirm the Python process is reading the VB-CABLE input device, not a microphone.
- Run **--list-audio-devices** again and re-check the device index after reboots or driver changes.

Wrong device selected

- Prefer **device_index** once the correct device is known.
- If the index changes often, remove **device_index** and rely on **device_name_contains**.

Sample-rate mismatch

- Keep SDR#, Windows device settings, and the detector config aligned at **48000 Hz**.
- WAV mode will fail fast if the file sample rate does not match the configured rate.

No detections despite a visible carrier

- Confirm SDR# is in **USB** mode.
- Confirm the carrier lands near the detector band **1200-1600 Hz**.
- Check that the tuned SDR# frequency still matches the expected GRAVES offset.
- Review whether the signal is present in **live_waterfall.png**.

Too many false positives

- Tighten detector thresholds in **[detector_profiles.graves_sdrsharp]**.
- Start with **trigger_db_above_baseline**, **peak_to_median_db_min**, and **max_near_peak_bins**.
- Validate changes first with **--input-wav** before trusting a live run.

Callback overflow or dropped audio

- Look for **WARNING:** lines in the console.
- Increase **queue_max_blocks**.
- Reduce background CPU load on the LattePanda Alpha.
- Avoid running heavy GUI tasks on the same machine during continuous monitoring.

Configuration Reference

[audio_inputs.sdrsharp_vb_cable]

- `device_index`: exact Windows input device index
- `device_name_contains`: fallback substring match
- `sample_rate`: expected audio sample rate
- `channels`: input channels from the Windows device
- `block_size`: detector block size
- `dtype`: `sounddevice` input sample format
- `queue_max_blocks`: backlog size before audio blocks are dropped

[runtime.local_sdrsharp]

- `output_dir`: root directory for logs and artifacts
- `log_level`: reserved runtime verbosity field
- `save_spectrogram`: enable candidate spectrogram PNGs
- `save_wav`: enable candidate WAV review files
- `save_detection_waterfall`: enable rolling waterfall and candidate snapshot output
- `waterfall_path`: rolling waterfall PNG path

[detector_profiles.graves_sdrsharp]

This holds the tuned meteor detection thresholds. Treat it as the algorithm profile, not as device config.